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Assignment 4: Vital Sign Description
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Vital Sign Description: Oxygen saturation (SpO_2)

Peripheral capillary oxygen saturation (SpO_2) is a vital sign that measures the percentage of hemoglobin in the blood that is carrying oxygen. Put simply, it indicates what percentage of your blood is saturated with oxygen. SpO_2 is an important indicator of how effectively oxygen is being transported from the lungs to the body's tissues and organs (1). Adequate oxygen delivery is essential for maintaining normal cell function and energy production, particularly in vital organs such as the brain, heart and kidneys. (2)

Oxygen saturation is most commonly measured using a pulse oximeter, a non-invasive device that clips onto a part of the body, usually a fingertip (1). **A normal SpO_2 reading in healthy individuals typically ranges from 95% to 100%** (3). Values below this range may indicate hypoxemia (<94% SpO_2), a condition in which there is low oxygen in the blood. Hypoxemia can lead to hypoxia (low oxygen in body tissues), meaning that insufficient oxygen is delivered to the body's cells, increasing the risk of organ dysfunction (3).

In triage, oxygen saturation is an important vital sign used to quickly assess a patient's respiratory status. Low SpO_2 readings can indicate impaired breathing or oxygen delivery and may increase a patient's priority level for urgent assessment and treatment to prevent deterioration. It is especially important because symptoms of hypoxemia may not always be obvious, they can develop suddenly or gradually over time. Overall, SpO_2 measurement is essential for healthcare professionals to identify patients at risk of respiratory compromise and help with immediate clinical decisions.

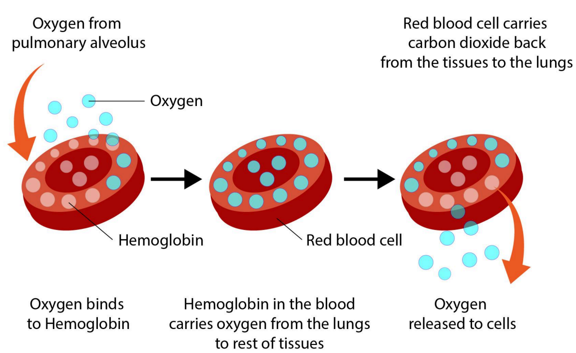

Figure 1: Simplified representation of the loading and unloading of a hemoglobin with oxygen.
Image source: **[Oxygen Saturation]**

Figure 2: Symptoms of Hypoxemia
(Low oxygen in the blood)
Image source: **hypoxemia**

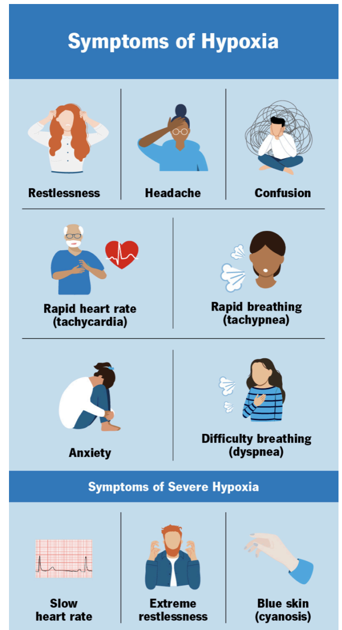

Figure 3: Symptoms of Hypoxia
(Low oxygen in the body tissues)
Image source: **hypoxia**

Reference:

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